

# INTRODUCTION TO FIREWALLD

GUIDED BY : VIPUL MISTRI

Q.1 Install firewalld on your Linux system, start the service, and verify its status using systemctl and firewall-cmd commands.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 20 18:39
root@localhost:/mnt

[root@localhost mnt]# dnf install firewalld
Last metadata expiration check: 2:12:36 ago on Thu 20 Aug 2026 04:26:37 PM IST.
Package firewalld-1.3.4-7.el9.noarch is already installed.
Dependencies resolved.

=====
Package Architecture Version Repository Size
=====
Upgrading:
firewalld noarch 1.3.4-20.el9 baseos 535 k
firewalld-filessystem noarch 1.3.4-20.el9 baseos 8.7 k
python3-firewall noarch 1.3.4-20.el9 baseos 393 k
=====
Transaction Summary
=====
Upgrade 3 Packages

Total download size: 936 k
Is this ok [y/N]: y
Downloading Packages:
(1/3): firewalld-filessystem-1.3.4-20.el9.noarch.rpm 15 kB/s | 8.7 kB 00:00
(2/3): python3-firewall-1.3.4-20.el9.noarch.rpm 296 kB/s | 393 kB 00:01
(3/3): firewalld-1.3.4-20.el9.noarch.rpm 398 kB/s | 535 kB 00:01
-----
Total 482 kB/s | 936 kB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Running scriptlet: firewalld-1.3.4-20.el9.noarch 1/1
Preparing : 1/1
Upgrading : python3-firewall-1.3.4-20.el9.noarch 1/6
Upgrading : firewalld-filessystem-1.3.4-20.el9.noarch 2/6
Upgrading : firewalld-1.3.4-20.el9.noarch 3/6
Running scriptlet: firewalld-1.3.4-20.el9.noarch 3/6
Running scriptlet: firewalld-1.3.4-7.el9.noarch 4/6
Cleanup : firewalld-1.3.4-7.el9.noarch 4/6
Running scriptlet: firewalld-1.3.4-7.el9.noarch 4/6
Cleanup : firewalld-filessystem-1.3.4-7.el9.noarch 5/6
Cleanup : python3-firewall-1.3.4-7.el9.noarch 6/6
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 20 18:39
root@localhost:/mnt

[root@localhost mnt]# systemctl status firewalld
Unit firewalld.service could not be found.
Unit firewalld.service could not be found.
[root@localhost mnt]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-20 18:39:18 IST; 26s ago
     Docs: man:firewalld(1)
    Main PID: 40083 (firewalld)
       Tasks: 2 (limit: 22799)
      Memory: 24.8M
         CPU: 278ms
    CGroup: /system.slice/firewalld.service
            └─40083 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

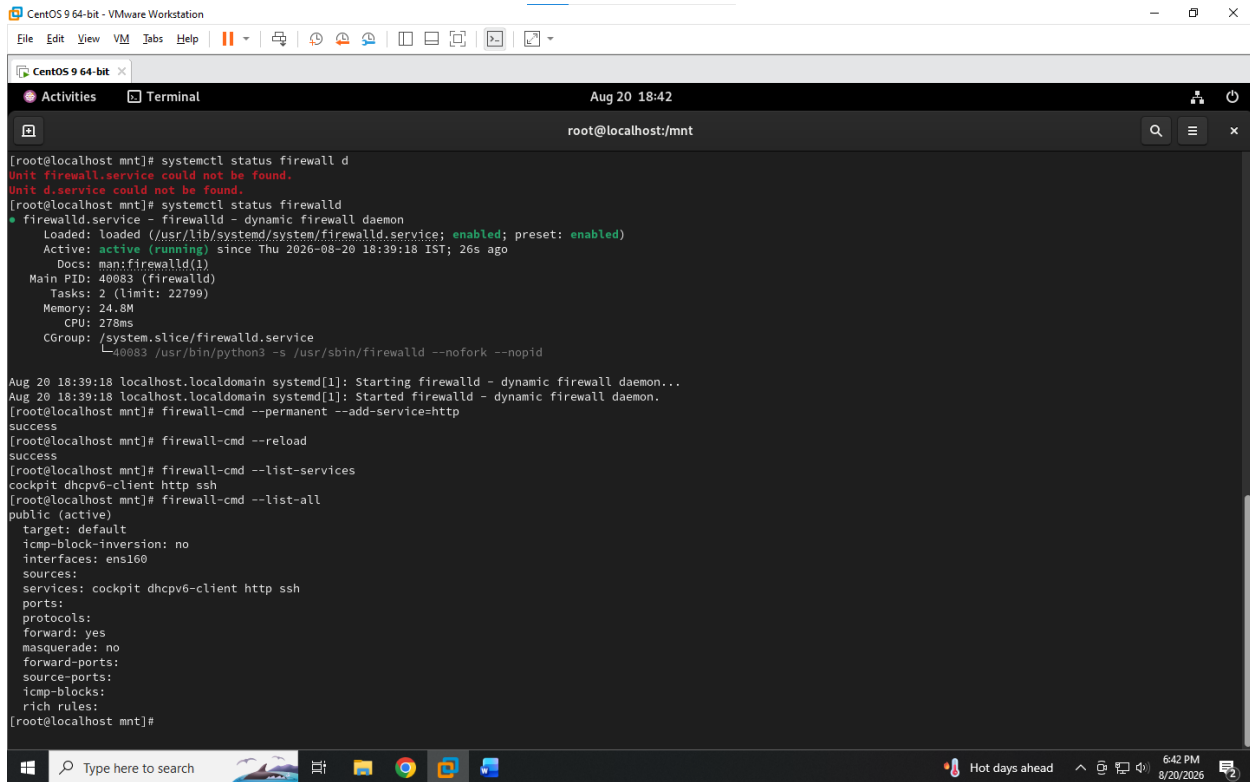
Aug 20 18:39:18 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon...
Aug 20 18:39:18 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
[root@localhost mnt]#
```

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Q.2 Add a new rule to firewalld to allow incoming HTTP (port 80) traffic, then use firewall-cmd to confirm the rule is active. Hint: Use the --permanent and --reload options to make changes persistent.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal Aug 20 18:42
root@localhost:/mnt

[root@localhost mnt]# systemctl status firewall d
Unit firewall.service could not be found.
Unit d.service could not be found.
[root@localhost mnt]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-20 18:39:18 IST; 26s ago
     Docs: man:firewalld(1)
    Main PID: 40083 (firewalld)
      Tasks: 2 (limit: 22799)
    Memory: 24.8M
       CPU: 278ms
    Group: /system.slice/firewalld.service
           └─40083 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

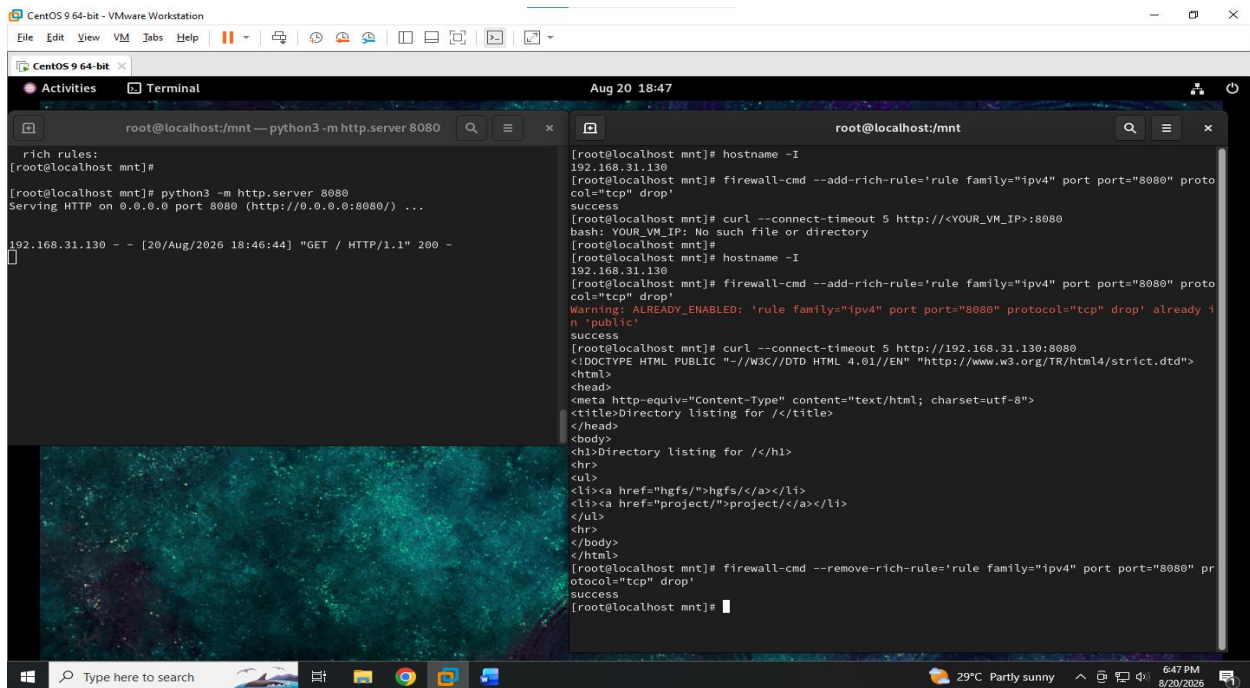
Aug 20 18:39:18 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon...
Aug 20 18:39:18 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
[root@localhost mnt]# firewall-cmd --permanent --add-service=http
success
[root@localhost mnt]# firewall-cmd --reload
success
[root@localhost mnt]# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
[root@localhost mnt]# firewall-cmd --list-all
public (active)
target: default
icmp-block-inversion: no
interfaces: ens160
sources:
services: cockpit dhcpv6-client http ssh
ports:
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
[root@localhost mnt]#
```

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Q.3 Temporarily block all incoming traffic on port 8080 using firewalld, then test by attempting to access a simple web server (e.g., Python's http.server) running on that port.

ANS :



The screenshot shows a CentOS 9 64-bit VM window with a terminal. The terminal is split into two panes. The left pane shows the setup of a Python HTTP server on port 8080. The right pane shows the configuration of a firewall rule to block traffic on port 8080, followed by a test using curl to access the server, which fails with a 'No such file or directory' error. The terminal output is as follows:

```
rich rules:
[root@localhost mnt]#
[root@localhost mnt]# python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...

192.168.31.130 - - [20/Aug/2026 18:46:44] "GET / HTTP/1.1" 200 -

```

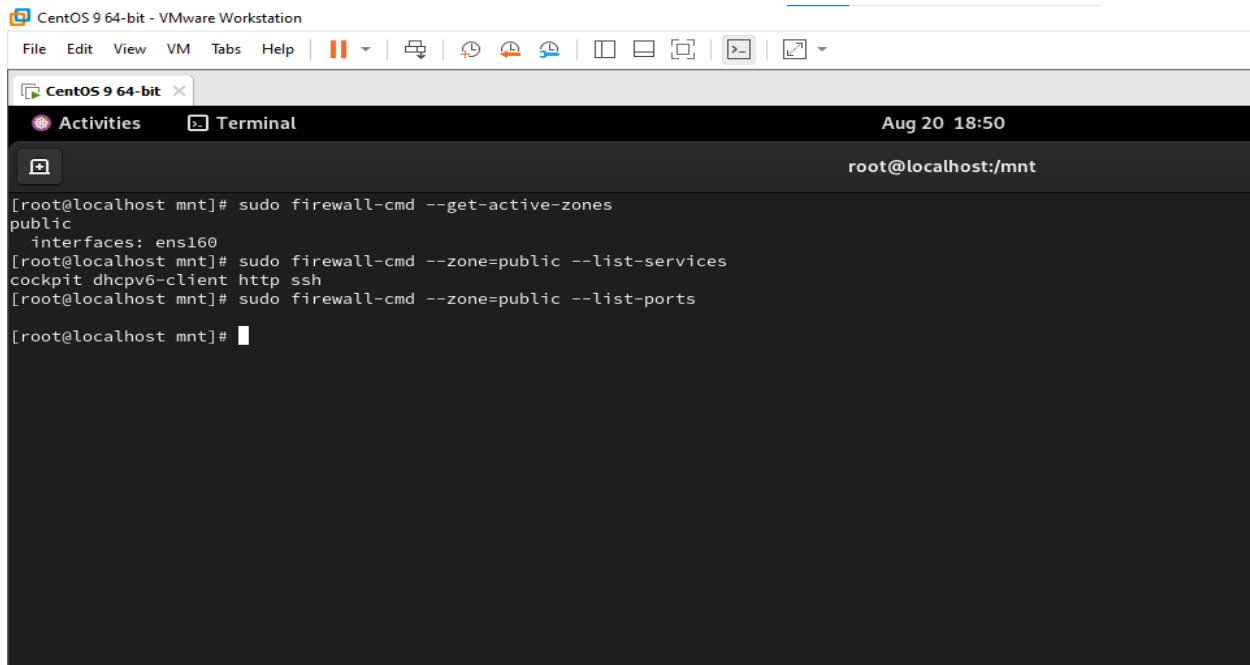
```
[root@localhost mnt]# hostname -I
192.168.31.130
[root@localhost mnt]# firewall-cmd --add-rich-rule='rule family="ipv4" port port="8080" protocol="tcp" drop'
success
[root@localhost mnt]# curl --connect-timeout 5 http://<YOUR_VM_IP>:8080
bash: YOUR_VM_IP: No such file or directory
[root@localhost mnt]#
[root@localhost mnt]# hostname -I
192.168.31.130
[root@localhost mnt]# firewall-cmd --add-rich-rule='rule family="ipv4" port port="8080" protocol="tcp" drop'
Warning: ALREADY_ENABLED: 'rule family="ipv4" port port="8080" protocol="tcp" drop' already in 'public'
success
[root@localhost mnt]# curl --connect-timeout 5 http://192.168.31.130:8080
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="hgfs/">hgfs/</a></li>
<li><a href="project/">project/</a></li>
</ul>
<hr>
</body>
</html>
[root@localhost mnt]# firewall-cmd --remove-rich-rule='rule family="ipv4" port port="8080" protocol="tcp" drop'
success
[root@localhost mnt]#
```

Q.4 List all currently active firewall zones and display the services and ports allowed in the public zone.

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ANS :

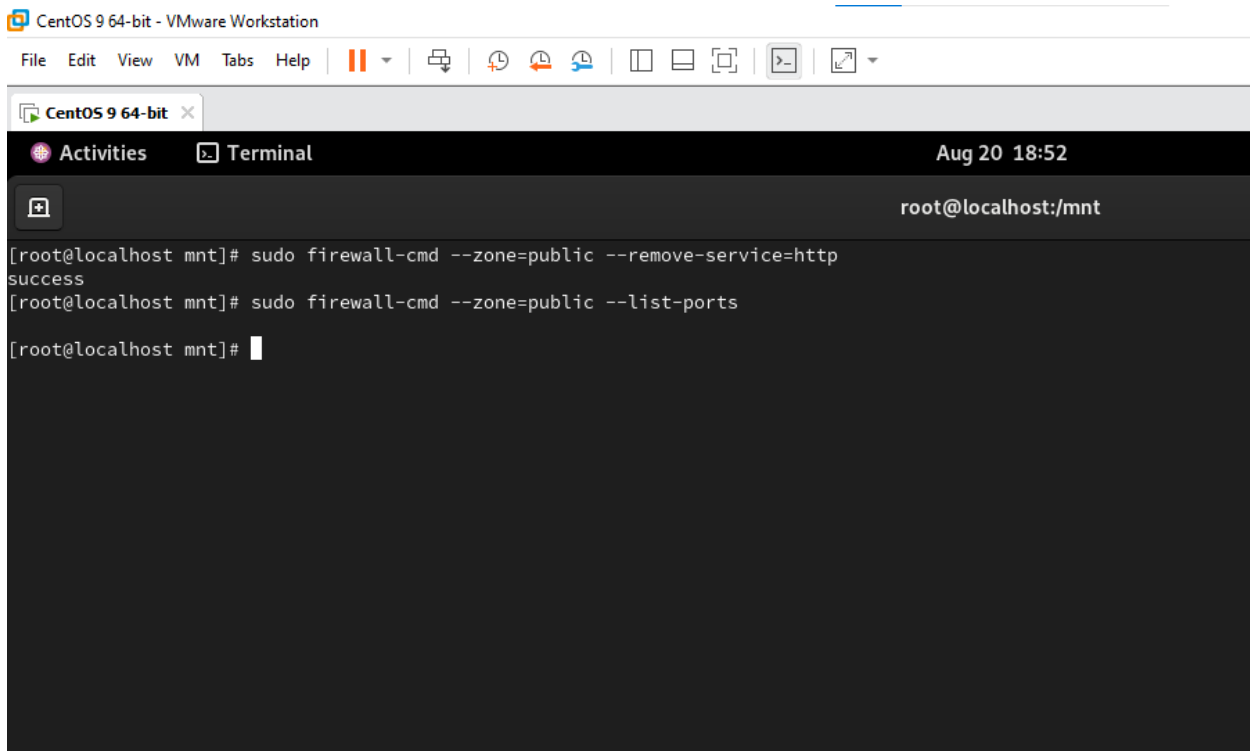


The screenshot shows a terminal window titled "CentOS 9 64-bit - VMware Workstation". The terminal is running the following commands:

```
[root@localhost mnt]# sudo firewall-cmd --get-active-zones
public
interfaces: ens160
[root@localhost mnt]# sudo firewall-cmd --zone=public --list-services
cockpit dhcpv6-client http ssh
[root@localhost mnt]# sudo firewall-cmd --zone=public --list-ports
[root@localhost mnt]#
```

Q.5 Remove a custom rule you added earlier (such as the HTTP port 80 rule) and verify that the change has taken effect using firewall-cmd.

ANS :



The screenshot shows a terminal window titled "CentOS 9 64-bit - VMware Workstation". The terminal is running the following commands:

```
[root@localhost mnt]# sudo firewall-cmd --zone=public --remove-service=http
success
[root@localhost mnt]# sudo firewall-cmd --zone=public --list-ports
[root@localhost mnt]#
```